Communicative efficiency develops with age

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It is a universal feature of languages to assign less linguistic material to more predictable or frequent meanings (Gibson et al., 2019; Haspelmath, 2021). The association between form length and meaning is argued to derive from speakers' bias for efficient communication, reflecting the need to balance competing pressures: minimizing production effort while maximizing understandability (Gibson et al., 2019; Zipf, 1949). An efficient trade-off between these two pressures involves producing less linguistic material whenever possible, i.e., when understandability is not compromised, and producing more linguistic material only when it is essential for being understood. Indeed, there is abundant evidence showing that speakers use language efficiently (for a review see Levshina & Moran, 2021). Specifically, speakers tend to increase their effort in response to three types of noise in the message transmission process that compromise their ability to be understood. These include environmental noise, the inherent properties of the message itself (e.g., when it is surprising or ambiguous), and perceived difficulties in listener comprehension (for a review see Pate & Goldwater, 2015). Yet, notably, all of this evidence is limited in a key way: it derives exclusively from research involving adults; It is not clear whether children's language use is also shaped by communicative efficiency. Investigating whether such a pressure is already present in children is important for understanding both the development of communicative behaviour and the respective roles of adults and children in shaping language structure – a heavily debated question (Raviv et al., 2025). Here, we investigate the developmental trajectory of communicative efficiency using a novel experimental paradigm. In a series of three studies, 216 children between the ages of 4 and 10 played a

communication game with a simulated interlocutor using visual icons: they had to tell the simulated interlocutor which action they should perform when meeting another character (kiss or hit). Crucially, both effort and understandability were manipulated in this communication game. To simulate effort, messages could vary in magnitude, with messages of greater magnitude taking more effort and time to produce. Understandability was simulated differently across the three studies, each corresponding to one of the three types of noise in the message transmission process outlined previously. In all instances, messages of greater magnitude were robust to the noise. For example, in the study simulating the perceived comprehension ability of the addressee, messages were sent during some communicative turns to an interlocutor whose visual access was partially blocked. In these instances, messages of greater magnitude were successfully understood by the interlocutor, while those of lesser magnitude were not.

If children are communicatively efficient, then messages of greater magnitude should be used in the presence of noise, while messages of smaller magnitude should be used when noise is absent. Results show that across all studies children are communicatively efficient, yet this efficiency increases with age. We find a developmental increase in both the tendency to maximize understandability and the tendency to minimize effort (see Figure 1). Additionally, we found that the kind of noise affected children's behaviour differently. We discuss the implications of our results for theories of language evolution and change as well as cognitive development.

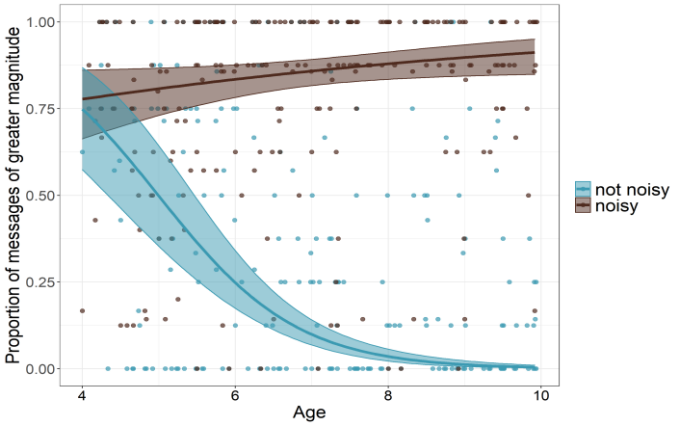

Figure 1: Proportion of messages of greater magnitude as function of age and noise, aggregating data across all three studies. Individual points represent by-participant means. Thick lines show estimated regression lines for noisy and not noisy trials, along with 95% confidence intervals based on the model's standard error.

References

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